

MUTATION STRATEGIES

Guide to Evolving Agent Genomes

Overview

Mutation is the engine of evolution. This guide covers strategies for mutating agent genomes effectively.

Mutation Types

Prompt Mutation

Modify system prompts while preserving intent.

Methods:

- Rephrase
- Elaborate
- Simplify
- Specialize

Config Mutation

Adjust parameters and settings.

Methods:

- Temperature ± 0.1
- Max tokens ± 100
- Top-p adjustment

Code Mutation

Modify agent source code.

Methods:

- Add conditionals
- Modify expressions
- Add error handling

Architecture Mutation

Change agent structure.

Methods:

- Add/remove methods
- Change inheritance
- Modify interfaces

Mutation Rates

Rate	Value	Use Case
Conservative	0.01-0.05	Fine-tuning near-optimal agents
Standard	0.05-0.15	General exploration
Aggressive	0.15-0.30	Escaping local optima
Extreme	0.30+	Complete restart/exploration

Best Practices

1. **Start conservative** — High mutation destroys good genomes
2. **Increase on plateaus** — If fitness stagnates, try higher rates
3. **Use adaptive rates** — Decrease mutation as fitness improves

4. **Preserve champions** — Never mutate your best agents (elitism)
5. **Log everything** — Flight Recorder tracks all mutations

MUTATION STRATEGIES | The Art of Change